Tail Rotor Dynamics for an RC Helicopter

Thrust, Induced Inflow, and Nonlinear Lead Compensation

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Abstract

This note develops a control-oriented model of an RC helicopter tail rotor. Starting from blade-element momentum theory (BEMT), we derive the thrust-versus-pitch relation, add dynamic inflow via the Pitt–Peters first-order model, invert the coupled system to obtain the blade pitch required to track a commanded thrust, and then extend the inverse to handle the static and dynamic nonlinearities of inflow softening. The result is a nonlinear lead compensator suitable for a tail-rotor inner loop. Realistic parameter ranges for RC tail rotors are tabulated.

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Nomenclature

The notation used throughout the note is collected below. Overbars denote trim (equilibrium) values; $\delta(\cdot)$ denotes small perturbations about trim; a superscript "cmd" denotes a commanded (reference) quantity. SI units are used unless otherwise stated; angles are in radians except where a table lists degrees.

Rotor geometry and operating condition

Symbol	Description	Units
R_t	Tail-rotor radius	m
r	Radial coordinate of a blade element, $0 \le r \le R_t$	m
c	Blade chord (assumed rectangular blade)	m
N_b	Number of blades	–
A_t	Rotor-disc area, $A_t = \pi R_t^2$	m^2
σ	Rotor solidity, $\sigma = N_b c/(\pi R_t)$	–
Ω_t	Rotor angular speed	$\mathrm{rad\,s^{-1}}$
$\Omega_t R_t$	Blade tip speed	$\mathrm{m\,s^{-1}}$
ρ	Air density (taken as $1.225\,\mathrm{kg\,m^{-3}}$)	$\mathrm{kg\,m^{-3}}$
Re	Blade-section Reynolds number	–

Blade-element aerodynamics

Symbol	Description	Units
θ_t	Blade collective pitch angle (control input)	rad
U_T	Tangential component of local flow at a blade section	$\mathrm{m\,s^{-1}}$
U_P	Perpendicular (axial) component of local flow	$\mathrm{m\,s^{-1}}$
ϕ	Local inflow angle, $\phi \approx U_P/U_T$	rad
α	Effective angle of attack, $\alpha = \theta_t - \phi$	rad
a	Two-dimensional lift-curve slope ($\approx 5.7\,\mathrm{rad^{-1}}$)	$\mathrm{rad^{-1}}$
C_ℓ	Section lift coefficient, $C_\ell = a\,\alpha$	–

Inflow and free-stream velocities

Symbol	Description	Units
v_i	Uniform induced velocity through the rotor disc (state)	$\mathrm{m\,s^{-1}}$
$v_{i,\mathrm{ss}}$	Steady-state value of v_i for the current operating point	$\mathrm{m\,s^{-1}}$
v_h	Hover induced velocity, $v_h = \sqrt{T_t/(2\rho A_t)}$	$\mathrm{m\,s^{-1}}$
V_∞	Axial component of the free-stream velocity through the rotor	$\mathrm{m\,s^{-1}}$
$V_\perp$	In-plane component of the free-stream velocity	$\mathrm{m\,s^{-1}}$
μ_z	Non-dimensional axial inflow, $\mu_z = V_\infty/(\Omega_t R_t)$	–
μ	Advance ratio (in-plane), $\mu = V_\perp/(\Omega_t R_t)$	–

Thrust and non-dimensional coefficients

Symbol	Description	Units
T_t	Tail-rotor thrust	N
C_T	Thrust coefficient, $C_T = T_t/[\rho A_t(\Omega_t R_t)^2]$	–
λ	Total inflow ratio, $\lambda = (v_i + V_\infty)/(\Omega_t R_t)$	–
λ_i	Induced-inflow ratio, $\lambda_i = v_i/(\Omega_t R_t)$	–
λ_h	Hover inflow ratio, $\lambda_h = \sqrt{C_T/2}$	–
λ_0	Uniform component of inflow in three-state Pitt–Peters	–
λ_s, λ_c	Longitudinal/lateral harmonic inflow components	–
C_{Mx}, C_{My}	Roll and pitch aerodynamic-moment coefficients on the disc	–

Dynamic inflow model

Symbol	Description	Units
τ_i	First-order inflow time constant (Pitt–Peters)	s
0.849	Apparent-mass coefficient, $8/(3\pi)$, in τ_i	–
$[M]$	Apparent-mass matrix in three-state Pitt–Peters	–
$[L]$	Inflow-gain matrix in three-state Pitt–Peters	–

Trim (equilibrium) quantities

Symbol	Description	Units
$\bar{T}$	Trim thrust (typically hover)	N
$\bar{v}_i$	Trim induced velocity	$\mathrm{m\,s^{-1}}$
$\bar{\theta}$	Trim blade pitch	rad
$\bar{\lambda}$	Trim inflow ratio	–
$\bar{C}_T$	Trim thrust coefficient	–
$\bar{\tau}_i$	Trim inflow time constant	s

Linearisation, plant model and compensator

Symbol	Description	Units
$\delta(\cdot)$	Small perturbation of a quantity about its trim value	(varies)
s	Laplace variable	s^{-1}
t	Time	s
K_1	Pitch-to-thrust gain, eq. (11)	$\mathrm{N\,rad}^{-1}$
K_2	Inflow-to-thrust gain, eq. (11)	$\mathrm{N\,s\,m}^{-1}$
k_v	Linearised sensitivity $\partial v_{i,\mathrm{ss}}/\partial T$, eq. (29)	$\mathrm{m\,s^{-1}\,N^{-1}}$
$K_2 k_v$	Dimensionless DC inflow-softening coupling	–
β	Softening factor, $\beta = 1 + K_2 k_v$	–
$S(\lambda)$	Self-consistent BEMT softening factor, eq. (45)	–
x	Internal state of the lead compensator	rad

Commanded (reference) quantities

Symbol	Description	Units
T_t^{cmd}	Commanded tail-rotor thrust	N
C_T^{cmd}	Commanded thrust coefficient	–
λ^{cmd}	Inflow ratio at the commanded trim, $\lambda^{\mathrm{cmd}} = \sqrt{C_T^{\mathrm{cmd}}/2}$	–
$\theta_t^{\mathrm{ss,cmd}}$	Steady-state pitch corresponding to C_T^{cmd}, eq. (51)	rad

Acronyms

BEMT	Blade Element Momentum Theory
LPV	Linear Parameter-Varying
DC	Direct current (steady-state, low-frequency limit)
TF	Transfer function
RC	Radio-controlled

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1 Introduction

The tail rotor of a single-main-rotor helicopter is, in steady flight, a near-quasi-steady thrust generator: its job is to balance the anti-torque of the main rotor and, via the pilot's pedal input, to provide directional control. On a radio-controlled (RC) model helicopter the same physics applies, but the rotor operates at much lower Reynolds number, much higher rotational speed, and with a single collective control. From a flight-control perspective, the relevant question is how the commanded blade pitch θ_t maps to the thrust T_t actually produced, in both steady and transient conditions.

Two effects make this map non-trivial. First, the thrust produced at a given pitch depends on the induced velocity v_i through the rotor: as the rotor generates lift, it accelerates air downwards through the disc, which changes the local angle of attack of every blade element and reduces the effective incidence. Second, the induced velocity itself takes a finite time to develop, because the wake has inertia; the relationship between v_i and T_t is therefore dynamic, not algebraic. The combined effect is a *lead–lag* response in thrust to a step in pitch: an immediate transient at the open-loop (no-inflow) gain, followed by partial relaxation to a lower steady value as inflow builds up.

This note develops a compact, control-oriented model of these effects. Section 1 derives the blade-element thrust relation, an expression linear in pitch and in induced velocity. Section 2 closes the loop on v_i using momentum theory and the first-order Pitt–Peters dynamic-inflow model. Section 3 inverts the coupled system to obtain the pitch required to track a thrust command and derives the linearised lead–lag transfer function. Section 4 shows how the gain and time constant of this plant vary with the trim point and quantifies the resulting *inflow-softening factor*. Section 5 presents three forms of nonlinear lead compensator—an LPV scheduled lead, an exact static inverse, and a two-point interpolant—and compares their behaviour. Section 6 tabulates parameter ranges for typical RC-helicopter classes and works a representative numerical example.

2 Tail Rotor Thrust vs. Blade Pitch

For a tail rotor with collective pitch θ_t (no cyclic), we use *Blade Element Momentum Theory* (BEMT). BEMT combines two independent views of the same rotor. The *local* (blade-element) view treats each spanwise strip of the blade as a two-dimensional aerofoil that produces lift according to its own angle of attack; integrating across the span and around the disc gives total thrust as a function of *both* the geometric pitch θ_t and the induced velocity v_i flowing through the disc. The *global* (momentum) view treats the disc as a whole and uses conservation of momentum on the air column passing through it to fix v_i given the thrust it is producing. The two views share v_i as the coupling variable and must be solved together. This section develops the local half; the global half is taken up in Section 3.

2.1 Blade Element Theory

Consider a single blade element—a thin spanwise strip—at radial station r on a rectangular blade of chord c, rotating at angular speed Ω_t. Two velocity components matter at this element:

$$U_T = \Omega_t\, r \qquad \text{(tangential, in-plane),} \tag{1}$$

$$U_P = v_i + V_\infty \qquad \text{(perpendicular, through-disc).} \tag{2}$$

U_T is simply the speed the element sees from rotation: zero at the hub, linear in r, peaking at $\Omega_t R_t$ at the tip. U_P is the *through-disc* flow—air being drawn or pushed perpendicular to the disc plane. It has two contributions: the induced velocity v_i produced by the rotor itself (always present whenever the rotor is producing thrust), and an external axial component V_∞ from translation of the airframe along the rotor shaft (for a tail rotor on a helicopter in straight flight, this is sideways translation; in hover, $V_\infty = 0$). The two notations V_∞ and $\mu_z \Omega_t R_t$ refer to the same quantity, with μ_z its non-dimensional form.

The element therefore sees an apparent wind whose direction is tilted out of the disc plane by the *inflow angle*

$$\phi = \tan^{-1}\left(\frac{U_P}{U_T}\right) \approx \frac{U_P}{U_T}, \tag{3}$$

where the small-angle approximation holds away from the root since $U_P \ll U_T$ over most of the span (typical values: $U_P \sim$ few m/s, $U_T \sim$ tens of m/s).

The blade is set at geometric pitch θ_t relative to the disc plane, but because the apparent wind is tilted by ϕ, the element's effective angle of attack is

$$\alpha = \theta_t - \phi. \tag{4}$$

This is the central mechanism of the entire section: θ_t is what we command, but induced inflow *steals incidence* from the blade by increasing ϕ, and only the difference α generates lift. The thrust loss that appears in the integrated result is the cumulative effect of this stolen incidence across the span.

Assuming a linear two-dimensional lift curve, $C_\ell = a\,\alpha$ with slope $a \approx 5.7\,\mathrm{rad}^{-1}$, the elemental thrust per blade is

$$dT = \tfrac{1}{2}\rho\,(U_T^2 + U_P^2)\,c\,C_\ell\,dr \approx \tfrac{1}{2}\rho\,(\Omega_t r)^2\,c\,a\,(\theta_t - \phi)\,dr, \tag{5}$$

where in the second form we have dropped U_P^2 against U_T^2 (consistent with ϕ small) and projected the section lift onto the disc normal ($\cos\phi \approx 1$). Section drag contributes only at second order in ϕ and is ignored here.

2.2 Integrated Thrust (Uniform Inflow)

Total rotor thrust is obtained by summing the elemental contribution over the span of one blade and multiplying by the number of blades N_b:

$$T_t \;=\; N_b \int_0^{R_t} dT \;=\; \tfrac{1}{2}\rho\,a\,c\,N_b\,\Omega_t^2 \int_0^{R_t} (\theta_t - \phi)\,r^2\,dr. \tag{6}$$

Assuming *uniform* induced velocity (v_i constant across the disc, a standard BEMT simplification), ϕ varies with r only through its denominator: $\phi(r) = U_P/(\Omega_t r) = (v_i + V_\infty)/(\Omega_t r)$. The integrand splits into two pieces,

$$\int_0^{R_t} \theta_t\,r^2\,dr = \frac{\theta_t R_t^3}{3}, \qquad \int_0^{R_t} \phi(r)\,r^2\,dr = \frac{v_i + V_\infty}{\Omega_t}\,\frac{R_t^2}{2}, \tag{7}$$

each weighted toward the tip—the first as r^2 (pitch contribution), the second as r (inflow loss). Non-dimensionalising by $\rho A_t(\Omega_t R_t)^2$ and introducing the rotor solidity $\sigma = N_b c/(\pi R_t)$ and the inflow ratio $\lambda = (v_i + V_\infty)/(\Omega_t R_t)$ gives the standard BEMT result:

$$\boxed{C_T \;=\; \frac{\sigma a}{2}\left(\frac{\theta_t}{3} - \frac{\lambda}{2}\right).} \tag{8}$$

The factor $\frac{1}{3}$ on θ_t reflects the cubic radial weighting of the pitch contribution; the factor $\frac{1}{2}$ on λ reflects the quadratic weighting of the inflow loss. The dimensional thrust is recovered as

$$T_t = C_T\,\rho\,A_t\,(\Omega_t R_t)^2, \qquad A_t = \pi R_t^2. \tag{9}$$

For a small RC tail rotor operating at essentially constant Ω_t (governor-controlled), (8) becomes linear in its two arguments θ_t and v_i:

$$T_t \approx K_1\,\theta_t - K_2\,v_i, \tag{10}$$

with (specialising to hover, $V_\infty = 0$)

$$K_1 = \frac{\rho A_t(\Omega_t R_t)^2\,\sigma a}{6}, \qquad K_2 = \frac{\rho A_t\,\Omega_t R_t\,\sigma a}{4}. \tag{11}$$

Physical interpretation of K_1 and K_2. K_1 is the thrust the rotor would produce per radian of collective if no induced inflow developed: the "open-loop" or "frozen-wake" pitch effectiveness. It is what the rotor delivers in the first instant after a pitch step, before the wake has had time to respond. K_2 is the rate at which thrust is given back to the induced wake as that wake builds up; it quantifies how much thrust each m/s of induced velocity costs us. Both constants scale with the common factor $\rho A_t \sigma a$, but they differ in their tip-speed dependence: K_1 carries $(\Omega_t R_t)^2$ while K_2 carries only $\Omega_t R_t$. Their ratio is purely kinematic,

$$\frac{K_2}{K_1} = \frac{3}{2\,\Omega_t R_t}, \tag{12}$$

independent of blade geometry, density, or lift-curve slope. This ratio sets the scale of the inflow-softening effect quantified in Section 5: faster-spinning rotors retain a larger fraction of their open-loop thrust gain.

Assumptions in force. The derivation above relies on the following simplifications, each of which is revisited in Section 7:

- Uniform induced velocity v_i across the disc (no spanwise variation, no harmonic components).
- Small inflow angle, $\phi \ll 1$, justifying $\tan\phi \approx \phi$, $\cos\phi \approx 1$, and $U_P^2 \ll U_T^2$.
- Linear two-dimensional lift curve up to stall, with no Reynolds- or Mach-number correction.
- Rectangular, untwisted blade with constant chord c.
- No tip loss and no root cutout (integration from 0 to R_t).
- Thrust contribution from drag is ignored ($O(\phi^2)$).
- Constant rotor speed Ω_t (held by the main-rotor governor).

The first assumption is the one momentum theory must subsequently repair, by supplying the v_i that closes (8).

3 Induced Inflow: Momentum Theory

The blade-element derivation of Section 1 produced $C_T = (\sigma a/2)(\theta_t/3 - \lambda/2)$, which contains *two* unknowns: the pitch θ_t (which we set) and the inflow ratio λ (which we do not). To close the system we need a second relationship between thrust and induced velocity. *Momentum theory* supplies it by treating the rotor disc as a whole and applying conservation of mass and momentum to the column of air passing through it.

3.1 The Actuator Disc and the Mass-Flow Argument

Replace the physical rotor by an idealised *actuator disc*: an infinitesimally thin permeable surface of area A_t that imparts a uniform pressure jump Δp to the air passing through it, without imparting swirl or losses. Draw a streamtube enclosing the disc, extending from far upstream (station 1, where the flow is undisturbed) through the disc (station 2, just below it—or station 3 just above, depending on orientation) and out to far downstream (station 4, where pressure has returned to ambient). For a rotor in hover the upstream flow is at rest and the streamtube contracts as the air accelerates; for a rotor in axial climb the upstream flow has velocity V_∞.

Three relationships govern this column of air:

1. *Mass conservation.* The mass flow rate $\dot{m} = \rho A_t U$ is the same at every station, where U is the local axial velocity. At the disc, $U = V_\infty + v_i$, where v_i is the induced velocity *at the disc.*
2. *Momentum.* The thrust on the rotor is the reaction to the change in axial momentum of the air, evaluated between far upstream and far downstream: $T_t = \dot{m}\,(w_\infty - V_\infty)$, where w_∞ is the axial velocity in the far wake.

3. *Bernoulli, separately above and below the disc.* Energy is conserved along streamlines on each side of the disc, but *not* across it (the rotor does work on the fluid). Applying Bernoulli from station 1 to just above the disc, and from just below the disc to station 4, and using the pressure jump Δp at the disc gives, after some algebra, the celebrated *Froude result*:

$$w_\infty \;=\; V_\infty + 2\,v_i. \tag{13}$$

The Froude result is non-obvious and worth dwelling on: *half* the total velocity change is acquired upstream of the disc, and the other half downstream. The wake therefore contracts to half the disc area and moves at twice the induced velocity. This factor of two appears in every momentum-theory thrust expression and is the reason rotors are *efficient* thrust generators at low disc loading.

Substituting $w_\infty - V_\infty = 2v_i$ into the momentum equation, together with $\dot{m} = \rho A_t(V_\infty + v_i)$, gives the *axial-flight* form of the momentum balance:

$$T_t \;=\; 2\,\rho\,A_t\,v_i\,(V_\infty + v_i). \tag{14}$$

3.2 Edgewise Flow: The Full Momentum Balance

For a tail rotor mounted on a translating airframe, the free-stream is not generally aligned with the rotor axis. Decompose the airframe velocity at the rotor into a component along the rotor axis (axial, V_∞, which feeds the streamtube directly) and a component in the disc plane (edgewise, $V_\perp$, which sweeps air sideways through the disc). The resultant velocity that the disc accelerates is the vector sum of the in-plane sweep $V_\perp$ and the through-disc flow $V_\infty + v_i$:

$$U_{\text{disc}} \;=\; \sqrt{(V_\infty + v_i)^2 + V_\perp^2}. \tag{15}$$

Glauert's hypothesis takes the mass-flow rate through the disc to be $\dot{m} = \rho A_t\,U_{\text{disc}}$, and the axial momentum imparted to be $2v_i$ (as in the purely axial case). The momentum equation then generalises to

$$\boxed{T_t \;=\; 2\,\rho\,A_t\,v_i\,\sqrt{(V_\infty + v_i)^2 + V_\perp^2}.} \tag{16}$$

The square-root factor is just the resultant velocity at the disc; the factor of two is Froude. In the purely axial case ($V_\perp = 0$) we recover (14), and in the hover limit ($V_\infty = V_\perp = 0$) the relation simplifies further to

$$T_t \;=\; 2\,\rho\,A_t\,v_i^2 \qquad\Longrightarrow\qquad v_h \;\equiv\; \sqrt{\frac{T_t}{2\,\rho\,A_t}}. \tag{17}$$

Reading the hover result. Equation (17) is the single most-cited result of rotor aerodynamics. It says induced velocity is set by *disc loading* T_t/A_t: rotors with larger disc area need less induced velocity for the same thrust, and induced power $T_t\,v_h \propto T_t^{3/2}/\sqrt{A_t}$ is lower as a result. A tail rotor (small A_t, high T_t/A_t) therefore has *high* v_i compared with a main rotor of equal thrust—a numerical example in Section 7 gives $v_h \sim 9\,\mathrm{m\,s^{-1}}$ for a typical 600-class tail rotor producing only 8 N of thrust.

3.3 Non-Dimensional Form

Dividing (16) through by $\rho A_t(\Omega_t R_t)^2$ converts it to a relation between non-dimensional groups. With the inflow ratios $\lambda_i = v_i/(\Omega_t R_t)$ and $\lambda = (V_\infty + v_i)/(\Omega_t R_t)$, and the edgewise advance ratio $\mu = V_\perp/(\Omega_t R_t)$, the result is

$$C_T \;=\; 2\,\lambda_i\,\sqrt{\mu^2 + \lambda^2} \qquad\Longleftrightarrow\qquad \lambda_i \;=\; \frac{C_T}{2\,\sqrt{\mu^2 + \lambda^2}}. \tag{18}$$

The square-root $\sqrt{\mu^2+\lambda^2}$ is the non-dimensional resultant velocity at the disc: μ is the in-plane (edgewise) component, λ the through-disc component, and their hypotenuse is what the disc actually accelerates.

Dimensional and non-dimensional forms in parallel. From here on we move between the dimensional induced velocity v_i and the inflow ratio λ_i freely; they encode the same physical quantity, related by $\lambda_i = v_i/(\Omega_t R_t)$. The dimensional form is convenient inside the dynamic-inflow ODE (Section 3.5), the non-dimensional form inside BEMT algebra, and the hover form v_h for sizing arguments.

3.4 Closing the Coupled System

Equation (18) and the BEMT result (8) are two relations between the same three quantities (θ_t, C_T, λ). For a given pitch θ_t they can be solved simultaneously for the steady-state inflow. In hover ($\mu = 0$ and $V_\infty = 0$, so $\lambda = \lambda_i$), substituting C_T from BEMT into the momentum result gives

$$\lambda_i \;=\; \frac{(\sigma a/2)(\theta_t/3-\lambda_i/2)}{2\lambda_i} \qquad\Longrightarrow\qquad \lambda_i^2+\frac{\sigma a}{8}\lambda_i-\frac{\sigma a}{12}\theta_t \;=\; 0. \tag{19}$$

This is a quadratic in λ_i whose positive root is the standard hover-BEMT inflow:

$$\boxed{\lambda_i \;=\; \frac{\sigma a}{16}\left[\sqrt{1+\frac{64\,\theta_t}{3\,\sigma a}}\;-\;1\right] \quad \text{(hover, } \mu=0\text{).}} \tag{20}$$

For small θ_t the bracket expands as $\frac{32\theta_t}{3\sigma a} - \mathcal{O}(\theta_t^2)$, giving $\lambda_i \approx 2\theta_t/3$—a useful order-of-magnitude check.

3.5 Why Steady Theory Is Not Enough: Wake Inertia

Equation (20) gives the *steady* inflow for a given pitch, but it ignores time. In reality the wake is a column of air with mass, and changing the rotor's thrust requires accelerating that column—which cannot happen instantaneously. If at time $t = 0$ the pilot steps the pitch up, the blade-element thrust jumps immediately to its open-loop value $K_1\,\delta\theta_t$ (the wake hasn't moved yet, so the local angle of attack is still $\theta_t - \phi$ at the old ϕ). Over the next few milliseconds the induced velocity builds toward its new steady value, the local angle of attack drops, and the thrust relaxes to its lower equilibrium. The rotor exhibits a *lead–lag* response in thrust because the inflow has *lag*.

For control-design purposes the simplest faithful model treats the uniform component of the induced velocity as a first-order state:

$$\boxed{\tau_i\,\dot v_i + v_i \;=\; v_{i,\mathrm{ss}},} \tag{21}$$

where $v_{i,\mathrm{ss}}$ is the steady-state induced velocity that solves (16) for the *current* thrust T_t, and τ_i is the wake-relaxation time constant. This is the *one-state Pitt–Peters* dynamic inflow model. Equation (21) is the inflow analogue of an RC low-pass filter: $v_{i,\mathrm{ss}}$ is the target, v_i is the state, and τ_i sets how quickly the state chases the target.

3.6 The Pitt–Peters Time Constant

What sets τ_i? Two mechanisms contribute. The first is the simple flow-traversal time: an air parcel takes roughly R_t/v_h to cross the rotor disc, so τ_i must scale this way on dimensional grounds alone. The second, more subtle effect is the *apparent mass* of the air column: when an impermeable disc is accelerated through fluid, the fluid in its immediate neighbourhood is accelerated with it, contributing an extra inertia even in the absence of a true rotor wake. Lamb's classical potential-flow result gives the apparent mass of a thin circular disc as $\rho\,(8/3)\,R_t^3$, a fraction $8/(3\pi) \approx 0.849$ of the mass of a sphere of radius R_t. Pitt and Peters identified this same coefficient as governing the lag of the uniform-inflow component, yielding

$$\boxed{\tau_i \;=\; \frac{8/(3\pi)}{4\,\Omega_t\,\lambda_i^{\mathrm{trim}}} \;=\; \frac{0.849\,R_t}{4\,v_h},} \tag{22}$$

where the two forms are equivalent via $\lambda_i^{\mathrm{trim}} = v_h/(\Omega_t R_t)$ at trim, and the substitution $\Omega_t\,\lambda_i^{\mathrm{trim}} R_t = v_h$ collapses the first form into the second. The right-hand form is the easier to interpret: τ_i is essentially the time for an air parcel travelling at the hover induced velocity to cross a quarter of the disc radius, scaled by the apparent-mass coefficient. Higher disc loading $\Rightarrow$ faster v_h $\Rightarrow$ shorter τ_i: hard-working rotors have snappier wakes.

Caveat on Pitt–Peters time constants. The closed-form (22) is known to underpredict measured inflow lag on full-scale rotors by a factor of 2–4. For RC tail rotors it likely underpredicts somewhat less, but the value should be regarded as a lower bound; see Section 7.

3.7 The Coupled Model

Combining the blade-element thrust relation with the dynamic-inflow state gives the complete tail-rotor model of this note:

$$\boxed{\begin{aligned} T_t &= \rho A_t(\Omega_t R_t)^2\,\frac{\sigma a}{2}\left(\frac{\theta_t}{3} - \frac{v_i + V_\infty}{2\Omega_t R_t}\right), && (23)\\ \dot{v}_i &= \frac{1}{\tau_i}\left(\frac{T_t}{2\rho A_t\sqrt{(V_\infty + v_i)^2 + V_\perp^2}} - v_i\right). && (24)\end{aligned}}$$

Equation (23) is BEMT, algebraic in θ_t and v_i. Equation (24) is Pitt–Peters: a first-order ODE in v_i driven by the instantaneous thrust through the momentum-theory steady-state map. The pair is a one-state nonlinear system with input θ_t and output T_t, parameterised by the flight condition $(V_\infty, V_\perp)$ and the rotor constants. Every subsequent section is either inverting this pair (Section 3), linearising it (Section 4), or designing a compensator around its linearisation (Section 5).

3.8 Three-State Pitt–Peters (Extension)

The one-state model captures only the *uniform* component of induced velocity. In forward flight the inflow becomes non-uniform: the advancing-side blade sees different inflow from the retreating-side blade, and a longitudinal gradient develops as the wake skews backward under $V_\perp$. Pitt and Peters captured these effects with two additional states—λ_s (longitudinal inflow harmonic, tilting the inflow front-to-back) and λ_c (lateral harmonic, tilting it side-to-side)—driven by the aerodynamic pitching and rolling moments on the disc:

$$[M]\begin{Bmatrix}\dot\lambda_0\\ \dot\lambda_s\\ \dot\lambda_c\end{Bmatrix} + [L]^{-1}\begin{Bmatrix}\lambda_0\\ \lambda_s\\ \lambda_c\end{Bmatrix} = \begin{Bmatrix}C_T\\ C_{Mx}\\ C_{My}\end{Bmatrix}, \tag{25}$$

with apparent-mass matrix $[M] = \mathrm{diag}\big(\frac{8}{3\pi}, \frac{16}{45\pi}, \frac{16}{45\pi}\big)$ and gain matrix $[L]$ depending on advance ratio and wake skew. The harmonic components matter for main rotors in forward flight and for cyclic-pitch control, but for a tail rotor in near-hover, with no cyclic and only modest edgewise flow from main-rotor wake, the uniform component dominates and the one-state model (21) is almost always sufficient. We use it for the remainder of this note.

4 Inverse Problem: Blade Pitch from Commanded Thrust

The coupled model of Section 3—blade-element thrust (23) and dynamic inflow (24)—maps a pitch input θ_t to a thrust output T_t. For control design we need to run this map *backwards*: given a commanded thrust $T_t^{\mathrm{cmd}}(t)$, what pitch $\theta_t(t)$ should we apply? The answer will sit inside the tail-rotor inner loop as a feedforward block: the outer loop (yaw control) produces a thrust demand, the inverse block translates it into pitch, and the servo drives the swashplate.

The difficulty is that the plant has *one input, one output, one state*—the state being the induced velocity v_i. The blade-element relation (23) is algebraic in (θ_t, v_i), but v_i is itself governed by a first-order ODE and so carries history. There are two routes through this: invert algebraically and propagate the state alongside (Section 4.1), or linearise about a trim point and work in the Laplace domain (Sections 4.2–4.5). Both have their place; the linearised form is what allows us to talk about *bandwidth*, *phase margin*, and *lead compensation* in the chapters that follow.

4.1 The Exact Instantaneous Inverse

The blade-element relation (10) is algebraically invertible: solving for θ_t gives

$$\boxed{\theta_t(t) \;=\; \frac{T_t^{\mathrm{cmd}}(t) + K_2\big(v_i(t) + V_\infty(t)\big)}{K_1}.} \tag{26}$$

This is the *exact* inverse at every instant, conditional on knowing the current v_i and V_∞: apply pitch (26) and the blade element will, instantaneously, produce thrust T_t^{cmd}.

Two practical issues prevent this from being the end of the story:

- v_i is rarely measured directly on an RC platform. It must be *estimated*, typically by integrating the Pitt–Peters ODE (21) in parallel with the controller, driven by the commanded thrust or by a thrust estimate.
- $V_\infty(t)$ varies with airframe motion (sideways translation for a tail rotor). Even when the rotor parameters are stationary, the external flow generally is not, and the inverse needs a reasonably current estimate of V_∞ to be accurate.

With these caveats, (26) together with the inflow ODE (21) *is* a complete control law—a non-linear dynamic inversion. Many practical tail-rotor controllers stop here. The remainder of the section develops the linearised form because it exposes the bandwidth structure of the plant and makes the lead-compensator design transparent.

4.2 Linearisation About Hover Trim

A linearised plant model lets us bring frequency-domain tools—transfer functions, poles and zeros, Bode plots—to bear on a controller that will ultimately run on the full nonlinear plant. The cost is that the linear model is only accurate near the chosen trim point.

Choosing trim. By *trim* we mean an equilibrium operating point of the plant: a constant pitch $\bar\theta$ producing a constant thrust $\bar T$ with a constant induced velocity $\bar v_i$, all satisfying both BEMT and momentum theory simultaneously. For a tail rotor on a hovering helicopter the natural choice is the trim that balances main-rotor anti-torque in still air. We specialise to hover ($V_\infty = V_\perp = 0$) because the algebra is cleanest and because the tail rotor spends most of its operating life near this condition; the non-hover case generalises by replacing $\bar v_i$ with the hover-equivalent $v_h = \sqrt{\bar T/(2\rho A_t)}$ and adjusting the trim accordingly.

Perturbation convention. Write each variable as a constant trim value plus a small perturbation:

$$\theta_t = \bar{\theta} + \delta\theta_t, \quad v_i = \bar{v}_i + \delta v_i, \quad T_t = \bar{T} + \delta T. \tag{27}$$

We retain terms linear in the δ-quantities and discard products of two perturbations (assumed small $\times$ small). The trim values themselves satisfy the steady model and cancel out of the perturbation equations, leaving a system in the δ-variables alone.

What needs linearising. The blade-element relation (10), $T_t = K_1\theta_t - K_2 v_i$, is *already linear* in its two arguments at fixed Ω_t; its perturbation form is simply

$$\delta T \;=\; K_1\,\delta\theta_t \;-\; K_2\,\delta v_i. \tag{28}$$

The nonlinearity is entirely in the momentum relation $\bar{T} = 2\rho A_t \bar{v}_i^2$, which couples thrust to induced velocity through a square. Differentiating implicitly,

$$\left.\frac{\partial T}{\partial v_i}\right|_{\text{trim}} \;=\; 4\,\rho A_t\,\bar{v}_i, \qquad \Longrightarrow \qquad \delta v_{i,\text{ss}} \;=\; \frac{1}{4\rho A_t \bar{v}_i}\,\delta T \;\equiv\; k_v\,\delta T, \tag{29}$$

where k_v is the linearised sensitivity of steady-state induced velocity to thrust at the chosen trim:

$$k_v \;\equiv\; \left.\frac{\partial v_{i,\text{ss}}}{\partial T}\right|_{\text{trim}} \;=\; \frac{1}{4\rho A_t \bar{v}_i} \;=\; \frac{\bar{v}_i}{2\bar{T}}, \qquad \text{units: m s}^{-1}\,\text{N}^{-1}. \tag{30}$$

The two forms come from the same expression via the hover relation $\bar{T} = 2\rho A_t \bar{v}_i^2$. Physically: k_v tells us *how much extra induced velocity each newton of additional thrust will pull through the disc, in steady state.* It is large for low-thrust trims (where $\bar{v}_i$ is small and any new thrust must accelerate a slow wake from near-rest) and small for high-thrust trims (where $\bar{v}_i$ is already substantial and incremental thrust matters less proportionally).

Linearised inflow dynamics. The Pitt–Peters ODE (21) is already linear in v_i at fixed τ_i. Substituting $v_{i,\text{ss}} = \bar{v}_i + k_v\,\delta T$ and subtracting the trim equation $\bar{v}_i = \bar{v}_i$ leaves

$$\tau_i\,\delta\dot{v}_i + \delta v_i \;=\; k_v\,\delta T. \tag{31}$$

Taking the Laplace transform with zero initial conditions ($\dot{(\cdot)} \to s\,(\cdot)$):

$$(\tau_i s + 1)\,\delta v_i(s) \;=\; k_v\,\delta T(s). \tag{32}$$

4.3 The Linearised Plant Transfer Function

Equations (28) and (32) are a two-equation linear system in the three Laplace-domain quantities $\delta\theta_t(s)$, $\delta T(s)$, $\delta v_i(s)$. Eliminate δv_i: from (32),

$$\delta v_i(s) \;=\; \frac{k_v}{\tau_i s + 1}\,\delta T(s). \tag{33}$$

Substitute into (28):

$$\delta T(s) \;=\; K_1\,\delta\theta_t(s) \;-\; K_2\,\frac{k_v}{\tau_i s + 1}\,\delta T(s). \tag{34}$$

Collect δT:

$$\delta T(s)\left[1 + \frac{K_2 k_v}{\tau_i s + 1}\right] \;=\; K_1\,\delta\theta_t(s), \tag{35}$$

and rearrange to standard form,

$$\boxed{\frac{\delta T(s)}{\delta\theta_t(s)} \;=\; \frac{K_1\,(\tau_i s + 1)}{\tau_i s + (1 + K_2 k_v)}.} \tag{36}$$

This is the linearised tail-rotor plant: thrust per radian of pitch perturbation about hover trim, valid for small $\delta\theta_t$ over a bandwidth limited by the linearisation.

4.4 Reading the Plant: Lead–Lag Structure

The plant (36) has one zero and one pole, both on the negative real axis:

$$\text{zero at } s = -\frac{1}{\tau_i}, \qquad \text{pole at } s = -\frac{1 + K_2 k_v}{\tau_i}. \tag{37}$$

Because $K_2 k_v > 0$, the pole sits at a higher frequency than the zero. The high-frequency and DC limits of the gain are

$$\lim_{s\to\infty} \frac{\delta T}{\delta\theta_t} = K_1 \qquad \text{(instantaneous, ``open-loop'' gain)}, \tag{38}$$

$$\lim_{s\to 0} \frac{\delta T}{\delta\theta_t} = \frac{K_1}{1 + K_2 k_v} \qquad \text{(steady-state, ``closed-wake'' gain)}. \tag{39}$$

The DC gain is lower than the instantaneous gain by a factor $1+K_2k_v$*.* A step in pitch produces an immediate thrust spike of size $K_1\,\delta\theta_t$ (before the wake responds), which then relaxes over a timescale $\tau_i/(1 + K_2k_v)$ to the lower steady value $K_1\,\delta\theta_t/(1 + K_2k_v)$. This is the *lead–lag* response described qualitatively in the introduction, now in closed form.

Physical reading of $K_2 k_v$. The dimensionless product K_2k_v measures how strongly induced inflow couples back into the thrust. Tracing its factors: K_2 is how much thrust each m/s of induced velocity steals; k_v is how much extra induced velocity each newton of thrust pulls through the disc. Their product closes the loop and gives the fractional thrust loss between the instantaneous and steady-state response:

$$\frac{\text{DC gain}}{\text{instantaneous gain}} = \frac{1}{1 + K_2 k_v}. \tag{40}$$

For typical RC tail rotors at hover trim, K_2k_v sits in the range 0.4–1.0—meaning the DC gain is anywhere from 50% to 70% of the open-loop gain. The thrust the rotor delivers in steady state is significantly less than what its blade pitch suggests. This phenomenon is *inflow softening*, and quantifying it across the operating envelope is the subject of Section 5.

Caveat: naive vs. self-consistent $K_2 k_v$. Computing K_2 from (11) and k_v from (29) *separately* and multiplying gives an answer that overstates the coupling, because the two factors are not independent at trim—they both depend on $\bar{v}_i$. A self-consistent expression, derived from the coupled BEMT–momentum equilibrium in Section 5, replaces the naive product with a simpler closed form. For now we use K_2k_v as a symbolic placeholder and refine its numerical value in the next chapter.

4.5 Inverting the Plant: The Lead Compensator

A feedforward compensator that cancels the plant dynamics is, by definition, the inverse of the plant. Reciprocating (36):

$$\boxed{\frac{\delta\theta_t(s)}{\delta T(s)} = \frac{\tau_i s + (1 + K_2 k_v)}{K_1\,(\tau_i s + 1)}.} \tag{41}$$

The roles of pole and zero are now swapped: the compensator has its *zero* where the plant had its *pole*, and vice versa. The compensator is in the classical sense a *lead network*: its zero sits at a lower frequency than its pole, so its phase contribution is positive across the intervening band. Cascading (41) → (36) gives the identity transfer function, so applying the compensator's output as pitch produces exactly the commanded thrust at every frequency—at least within the validity of the linearisation.

Time-domain reading. In the time domain, (41) responds to a thrust-command step by first applying an overdriving pitch transient ($\delta\theta_t \to \delta T/K_1 \cdot (1 + K_2 k_v)$ immediately—the high-frequency limit of the compensator), which then decays to the steady pitch $\delta T/K_1$ (the low-frequency limit). In words: the compensator anticipates the rotor's tendency to fall back as inflow builds, and pre-emptively over-pitches by the softening factor to compensate. This is the linearised version of the nonlinear lead introduced in Section 5 and refined in Section 6.

Limitations of the linear compensator. The compensator (41) has gain K_1, zero β/τ_i, and pole $1/\tau_i$, all of which depend on the trim point through $\bar{v}_i$ (and hence through $\bar{T}$). A single fixed compensator designed at hover will mis-track at other thrust levels by amounts quantified in Section 5. Section 6 develops three constructions—an LPV-scheduled lead, an exact nonlinear static inverse, and a two-point interpolant—that handle this variation across the operating envelope.

5 Inflow Softening and the Nonlinear Lead

5.1 What Inflow Softening Means

The DC gain $K_1/(1 + K_2 k_v)$ varies with the trim point because $k_v \propto 1/\bar{v}_i \propto 1/\sqrt{\bar{T}}$. The factor $(1 + K_2 k_v)$ is the *inflow softening factor*—softening is strongest at low thrust.

5.2 BEMT Derivation of the Softening Factor

In hover, the coupled equations are

$$C_T = \frac{\sigma a}{2}\left(\frac{\theta_t}{3} - \frac{\lambda}{2}\right), \qquad \text{(BEMT)}$$

$$\lambda = \sqrt{C_T/2}. \qquad \text{(Momentum)}$$

Eliminating C_T:

$$\lambda^2 = \frac{\sigma a}{4}\left(\frac{\theta_t}{3} - \frac{\lambda}{2}\right), \tag{42}$$

with closed form

$$\lambda(\theta_t) = \frac{\sigma a}{16}\left[\sqrt{1 + \frac{64\theta_t}{3\sigma a}} - 1\right]. \tag{43}$$

Trim sensitivity. Differentiating implicitly:

$$2\lambda\,\frac{d\lambda}{d\theta_t} = \frac{\sigma a}{4}\left(\frac{1}{3} - \frac{1}{2}\frac{d\lambda}{d\theta_t}\right) \quad\Longrightarrow\quad \frac{d\lambda}{d\theta_t} = \frac{\sigma a}{24\lambda + 3\sigma a/2}. \tag{44}$$

Then

$$\boxed{\left.\frac{dC_T}{d\theta_t}\right|_{\text{steady}} = \frac{\sigma a}{6}\cdot\underbrace{\frac{1}{1 + \dfrac{\sigma a}{16\lambda}}}_{S(\lambda)}.} \tag{45}$$

The factor $\sigma a/6$ is the instantaneous (high-frequency) gain; $S(\lambda)$ is the softening factor. Comparing with the linearised form $S = 1/(1 + K_2 k_v)$ from Section 4.4 identifies the self-consistent value of the coupling product as

$$\boxed{\left.K_2 k_v\right|_{\text{self-consistent}} = \frac{\sigma a}{16\,\lambda_h}, \qquad \lambda_h = \sqrt{C_T/2}.} \tag{46}$$

This is the expression to use in design: it bakes in the trim consistency between K_2 and k_v that a naive separate calculation would miss.

5.3 Magnitude Across the Envelope

For a 600-class tail rotor ($\sigma = 0.138$, $a = 5.7$, hover $\bar{C}_T = 0.024$, $\bar{\lambda} = 0.110$):

Thrust	C_T	λ	$K_2 k_v = \sigma a/(16\lambda)$	$S(\lambda)$
25% hover	0.006	0.055	0.89	0.53
50% hover	0.012	0.078	0.63	0.61
Hover	0.024	0.110	0.45	0.69
150% hover	0.036	0.135	0.36	0.73
200% hover	0.048	0.156	0.32	0.76

The DC gain varies by roughly 40% across the operating envelope. A fixed-gain lead compensator will mis-track by this amount at the extremes.

5.4 Time Constant Varies Too

The Pitt–Peters time constant $\tau_i = 0.849/(4\Omega_t\lambda)$ scales as $1/\lambda \propto 1/\sqrt{C_T}$. For the same 600-class rotor:

Thrust	τ_i [ms]
25% hover	5.5
Hover	2.7
200% hover	1.9

Both gain and time constant of the rotor depend on the trim operating point. Both must be scheduled.

6 Nonlinear Lead Compensator

6.1 Approach A: Gain Scheduling on Commanded Thrust (LPV)

Schedule the lead-compensator coefficients on the current trim λ^{cmd} computed from the commanded thrust:

$$\lambda^{\text{cmd}}(t) = \sqrt{\frac{C_T^{\text{cmd}}(t)}{2}} = \sqrt{\frac{T_t^{\text{cmd}}(t)}{2\rho A_t(\Omega_t R_t)^2}}. \tag{47}$$

The scheduled compensator is

$$\boxed{\begin{aligned} \tau_i(t) &= \frac{0.849}{4\,\Omega_t\,\lambda^{\text{cmd}}(t)}, \\ \beta(t) &\equiv 1 + K_2 k_v = 1 + \frac{\sigma a}{16\,\lambda^{\text{cmd}}(t)}, \\ \frac{\theta_t(s)}{T_t(s)} &= \frac{\tau_i(t)\,s + \beta(t)}{K_1\big(\tau_i(t)\,s + 1\big)}. \end{aligned}} \tag{48}$$

In continuous state form, with a single internal state x,

$$\dot{x} = -\frac{1}{\tau_i(t)}\,x + \frac{\beta(t) - 1}{\tau_i(t)\,K_1}\,T_t^{\text{cmd}}, \tag{49}$$

$$\theta_t = \frac{1}{K_1}\,T_t^{\text{cmd}} + x. \tag{50}$$

The $1/K_1$ feedforward gives the instantaneous inversion; the state x adds the low-frequency correction.

6.2 Approach B: Exact Static Inversion + Inflow Filter

Bypass linearisation. The steady-state map $\theta_t \leftrightarrow C_T$ is exact in closed form: from (8) with $\lambda = \sqrt{C_T/2}$,

$$\boxed{\theta_t^{\mathrm{ss,cmd}} = \frac{6\,C_T^{\mathrm{cmd}}}{\sigma a} + \frac{3\,\lambda^{\mathrm{cmd}}}{2}, \qquad \lambda^{\mathrm{cmd}} = \sqrt{C_T^{\mathrm{cmd}}/2}.} \tag{51}$$

This static inverse, applied directly as feedforward, intentionally over-drives the rotor for the duration of the inflow lag: the rotor produces a brief thrust spike that builds inflow up to λ^{cmd}, after which thrust settles to C_T^{cmd}. This is the nonlinear analogue of the lead compensator, and is structurally simpler than the LPV form: a single algebraic expression in C_T^{cmd}.

For completeness, an inflow state with scheduled time constant can be propagated alongside,

$$\dot{\lambda} = \frac{1}{\tau_i(\lambda^{\mathrm{cmd}})}\big(\lambda^{\mathrm{cmd}} - \lambda\big), \tag{52}$$

to expose the actual inflow as a state—useful for simulation, monitoring, or hybridising with feedback.

6.3 Approach C: Two-Point Approximation

For embedded implementations where LPV is too heavy, interpolate the compensator zero β/τ_i and pole $1/\tau_i$ linearly in $\sqrt{T_t^{\mathrm{cmd}}}$ between two design points (low and high thrust).

6.4 Comparison

Compensator	Initial θ_t	Settled θ_t	Notes
None ($1/K_1$ only)	correct	undershoots 31%	thrust slow to build
Fixed lead at hover	correct	DC tracks at hover	overshoots at high thrust
LPV lead (A)	correct	DC tracks everywhere	needs schedule
Exact static inverse (B)	exceeds $\theta^{\mathrm{ss,cmd}}$ by factor $\beta(\lambda^{\mathrm{cmd}})$ (intentional lead)	DC tracks exactly	cleanest, single expression

The static-inverse approach (B) is generally preferred: the nonlinearity falls out of the BEMT algebra without a schedule, and the dynamic correction is a single first-order state. The LPV form (A) is more rigorous for control analysis but rarely justifies its complexity in a tail-rotor inner loop.

6.5 Scheduling Variable Choice

Should the schedule key on commanded thrust T_t^{cmd} or on the actual (estimated) thrust T_t? Both have drawbacks. Scheduling on the command assumes the rotor is tracking; scheduling on the measured thrust is self-consistent but susceptible to noise. A practical compromise is to schedule on a low-pass filtered command, with cutoff near $1/\bar{\tau}_i$—this is "scheduling on slow variables" in LPV terminology and is sound provided the scheduling-variable bandwidth lies below the loop bandwidth.

7 Parameter Ranges for RC Tail Rotors

7.1 Geometric and Operating Parameters

Symbol	Quantity	Micro	450	550–700	800+
R_t	Tail radius [m]	0.03–0.05	0.06–0.08	0.10–0.13	0.14–0.17
c	Blade chord [m]	0.008–0.012	0.015–0.020	0.022–0.028	0.028–0.035
N_b	Number of blades	2	2	2 (occ. 4)	2 or 4
Ω_t	Rotor speed [$\mathrm{rad\,s^{-1}}$]	800–1500	600–1000	500–800	400–700
$\Omega_t R_t$	Tip speed [m/s]	40–60	50–75	70–100	80–110
σ	Solidity	0.10–0.16	0.12–0.18	0.12–0.18	0.12–0.20
a	Lift-curve slope [1/rad]	5.5–5.75	5.5–5.75	5.5–5.75	5.5–5.75

The lift-curve slope sits near $2\pi \approx 6.28$ in theory, reduced to roughly 5.7 by finite-aspect-ratio and tip-loss effects, and falls further at very low Reynolds ($Re < 50\,000$).

7.2 Aerodynamic Trim Values

Symbol	Quantity	Micro	450	550–700	800+
$\bar{T}$	Hover thrust [N]	0.3–1	1–4	4–12	12–30
$\bar{v}_i$	Hover induced vel. [m/s]	4–8	5–10	7–13	9–15
$\bar{\theta}$	Trim pitch [deg]	6–10	6–10	6–10	6–10
τ_i	Inflow time constant [ms]	4–10	8–20	15–35	25–60

7.3 Derived Coefficients

Using $\rho = 1.225\,\mathrm{kg\,m^{-3}}$:

Symbol	Quantity	Micro	450	550–700	800+
A_t	Disc area [$\mathrm{m^2}$]	0.003–0.008	0.011–0.020	0.031–0.053	0.062–0.091
K_1	Pitch gain [N/rad]	3–10	10–40	40–150	150–400
K_2	Inflow gain [N s/m]	0.3–1.0	0.8–2.5	2.5–8	8–20
k_v	Inflow sensitivity [m/(s N)]	0.5–2.0	0.3–1.0	0.2–0.6	0.15–0.4
$K_2 k_v$	DC softening factor	0.4–1.0	0.4–1.0	0.4–1.0	0.4–1.0

The dimensionless product $K_2 k_v$ is roughly the same across classes—a physical-similarity result, since both factors scale with similar combinations of ρA_t, $\Omega_t R_t$, and $\bar{v}_i$.

7.4 Worked Example: 600-class

$$
\begin{aligned}
R_t &= 0.115 \text{ m}, & c &= 0.025 \text{ m}, & N_b &= 2, \\
\Omega_t &= 700 \text{ rad/s}, & \Omega_t R_t &= 80.5 \text{ m/s}, & & \\
\sigma &= \tfrac{2\cdot 0.025}{\pi\cdot 0.115} = 0.138, & a &= 5.7, & & \\
\bar{T} &= 8 \text{ N}, & A_t &= 0.0415 \text{ m}^2. & &
\end{aligned}
$$

Hence

$$
\begin{aligned}
\bar{v}_i &= \sqrt{8/(2\cdot 1.225\cdot 0.0415)} = 8.9 \text{ m/s},\\
\tau_i &= 0.849\cdot 0.115/(4\cdot 8.9) \approx 2.7 \text{ ms (Pitt–Peters)},\\
K_1 &= \tfrac{1.225\cdot 0.0415\cdot 80.5^2\cdot 0.138\cdot 5.7}{6} = 43 \text{ N/rad},\\
K_2 &= \tfrac{1.225\cdot 0.0415\cdot 80.5\cdot 0.138\cdot 5.7}{4} = 8.1 \text{ N s/m},\\
k_v &= 1/(4\cdot 1.225\cdot 0.0415\cdot 8.9) = 0.55 \text{ m/(s N)},\\
\lambda_h &= \sqrt{0.024/2} = 0.110,\\
K_2 k_v\big|_{\text{consistent}} &= \frac{\sigma a}{16\lambda_h} = \frac{0.787}{1.76} = 0.45.
\end{aligned}
$$

7.5 Caveats

Pitt–Peters time constants are systematically low compared to flight-test data on full-scale rotors (factor of 2–4 underestimate); RC practitioners often use $\tau_i \approx 0.1$–0.3 rotor revolutions empirically. For design, the Pitt–Peters value should therefore be treated as a lower bound on the inflow lag, and either inflated by a fitted multiplier or verified against bench data before being used to set compensator coefficients. Blade lift-curve slope drops noticeably at low Reynolds numbers ($Re \sim 30\,000$–$80\,000$) typical of RC tail-rotor tips, especially with thin symmetric sections. Solidity calculations assume rectangular blades; tapered or paddle blades need an equivalent solidity weighted by $r^2 c(r)$. Finally, the tail rotor sits in the main-rotor wake, so $V_\perp$ in (16) is dominated by main-rotor inflow—the main source of tail-rotor effectiveness loss in turns.

8 Summary

The tail-rotor inner loop reduces to:

1. A near-linear blade-element thrust law, $T_t = K_1\theta_t - K_2 v_i$, with constants from (11).
2. A first-order inflow state, $\tau_i \dot{v}_i + v_i = v_{i,\text{ss}}(T_t)$, with τ_i from (22).
3. A coupled steady-state map $\theta_t \leftrightarrow C_T$ with closed form (43).
4. A self-consistent DC softening factor $\sigma a/(16\lambda_h)$ that replaces the naive product $K_2 k_v$.
5. A nonlinear feedforward inverse—either the LPV lead (41) scheduled on λ^{cmd}, or the exact static inverse (51)—that compensates for inflow softening across the full thrust envelope.